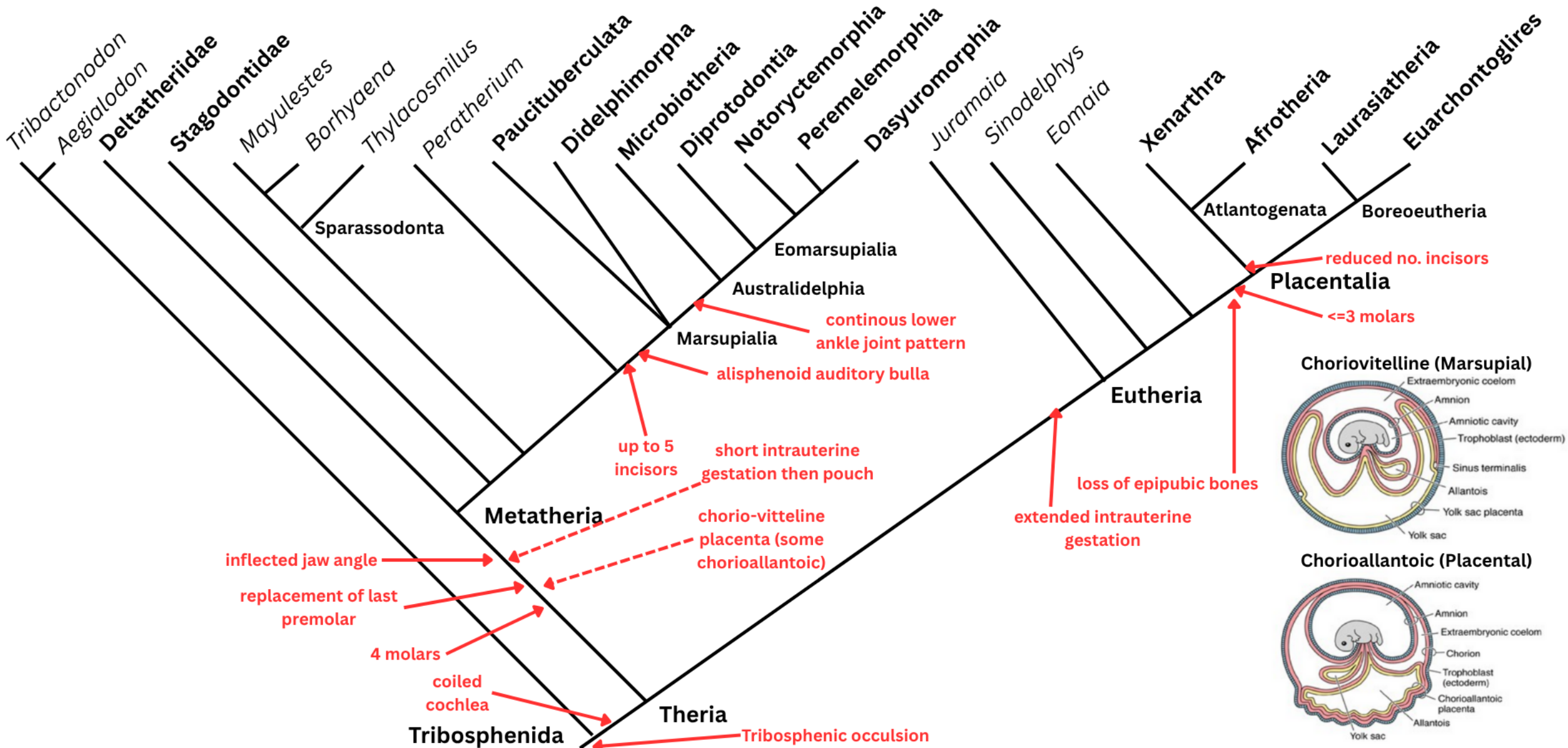
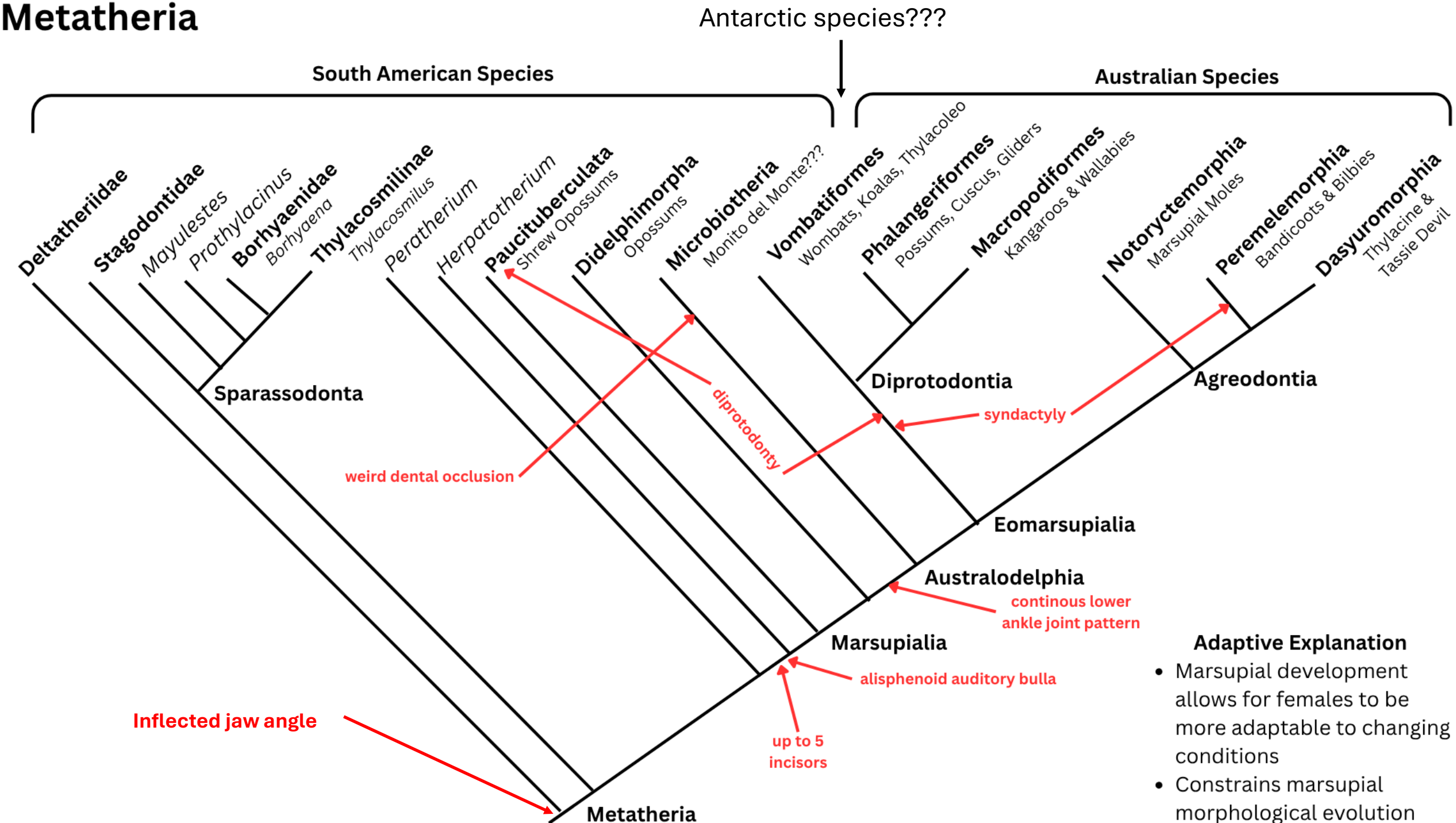
Theria


Tribactonodon
Aegialodon
Deltatheriidae
Stagodontidae
Mayulestes
Borhyaena
Thylacosmilus
Peratherium
Paucituberculata
Didelphimorpha
Microbiotheria
Diprotodontia
Notoryctemorphia
Peremelemorphia
Dasyuromorphia
Juramaia
Sinodelphys
Eomaia
Xenarthra
Afrotheria
Laurasiatheria
Euarchontoglires
Sparassodonta
Eomarsupialia
Australidelphia
Marsupialia
Atlantogenata
Boreoeutheria
reduced no. incisors
Placentalia
<=3 molars
continous lower ankle joint pattern
alisphenoid auditory bulla
Eutheria
up to 5 incisors
short intrauterine gestation then pouch
loss of epipubic bones
extended intrauterine gestation
Metatheria
chorio-vitteline placenta (some chorioallantoic)
inflected jaw angle
replacement of last premolar
4 molars
coiled cochlea
Theria
Tribosphenida
Tribosphenic occulsion
Choriovitelline (Marsupial)
Extraembryonic coelom
Amnion
Amniotic cavity
Trophoblast (ectoderm)
Sinus terminalis
Allantois
Yolk sac placenta
Yolk sac
Chorioallantoic (Placental)
Amniotic cavity
Amnion
Extraembryonic coelom
Chorion
Trophoblast (ectoderm)
Chorioallantoic placenta
Allantois
Yolk sac

Metatheria



Adaptive Explanation

- Marsupial development allows for females to be more adaptable to changing conditions
- Constrains marsupial morphological evolution

Key Marsupial Traits

- Inflected angle of jaw (reflected/inflected angular process)
- Epipubic bones (shared with monotremes)
- Alisphenoid auditory bulla (ecto/entotympanic in placentals)

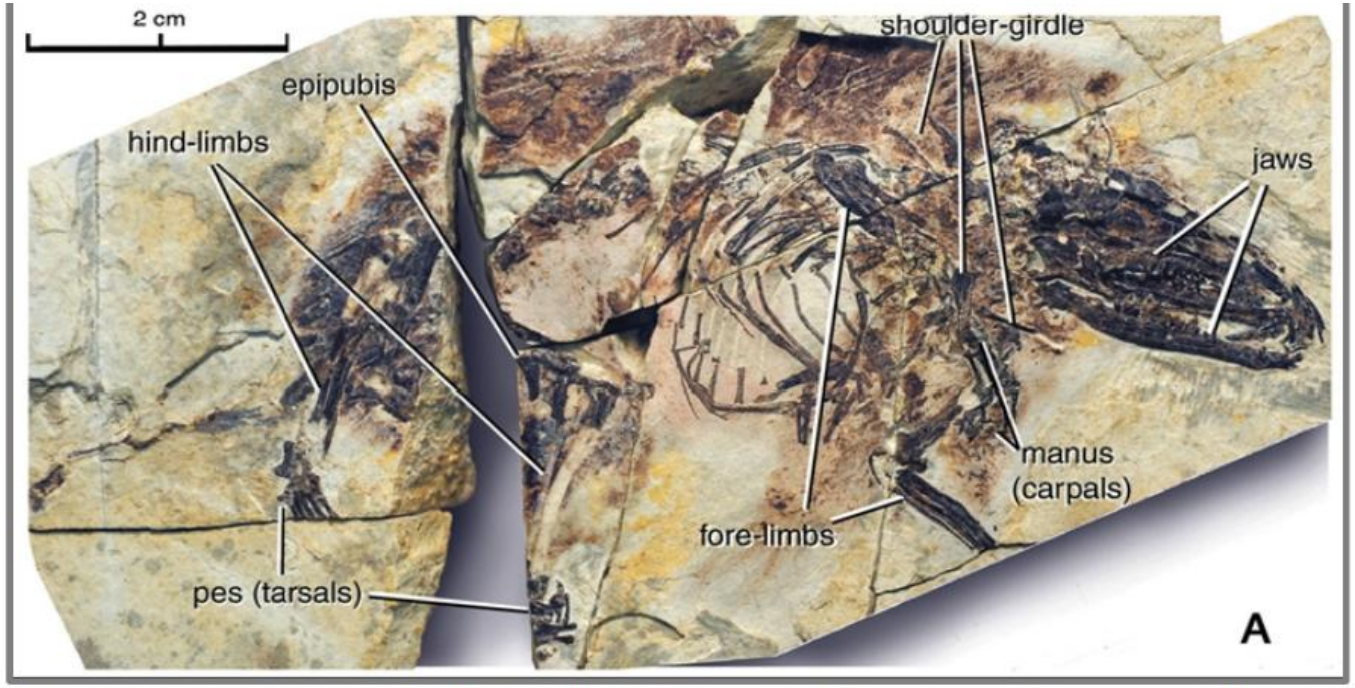
- one thing to note is various bones tend to evolve to cover the auditory bulla in placental mammals alongside the tympanic bones

- Replacement of ONLY the last premolar
- 4 Molars
- Up to 5 incisors
- Diastema between front teeth and antemolars
- Choriovitelline placenta (chorioallantoic in placentals)
- Chorioallantoic placenta in peramelians
- Unique ankle pattern in derived marsupials
- Supinator crest
- Broad navicular
- Oblique cuboid facet on calcaneum
- Large scaphoid, triquetrum, hamate

Oldest metatherian is probably deltatheridium based on revision of Sinodelphys in Bi et al. 2018.

Paleocene metatherians essentially had all anatomical hallmarks of marsupials except the alisphenoid bulla.

- Pucadelphys & Mayulestes



Sinodelphys szalayi, up until 2018 thought to be oldest metatherian (early Cretaceous, China), but is more likely eutherian (Bi et al. 2018 *Nature* 558:390-395)

Marsupial Diversity, Ecology, Evolution

South American Taxa:

Opossums – Racoon-sized, wide-spread generalists

Shrew-opossums – Small S. American insectivores

Monito del Monte – Small, Chilean and Argentinian faunivores

Australian Taxa:

Diprotodonts – Diverse in species, size, and niche – diprotodonty (two front teeth – like rodents)

- Thylacoleo was a predatory vombatiform from the Pleistocene
- Diprotodon was a giant wombat, similar niche to elephant

Peramelians – Range from 5kg to 100g, contain hoofed bandicoot

Notoryctemorphans – marsupial moles

Dasyuromorpha – Primarily predatory/carnivorous clade.

Metatherians:

Sparrasodonta – major clade of (hyper) carnivorous metatherians from south America

Borhyaena

Thylacosmilus

Overall Ecology and Evolution:

1. Metatherians evolve on southern continents (Gondwana) while eutherians evolve on northern.
2. Therefore, were a highly diverse group filling many niches.
3. S. America joining with N. America caused placentals to migrate into S. America, these outcompeted many marsupials.
4. Antarctica likely had a lineage of metatherians/marsupials (as it definitely had terror birds) but due to it moving towards the poles during the Cenozoic, this probably made these lineages go extinct.
5. Australia becomes the last surviving stronghold for marsupials due to its harsh climate and the advantages the chorio-vitelline placenta gives there.

†Diprotodon

https://www.instagram.com/p/CPSnI19AjNk/

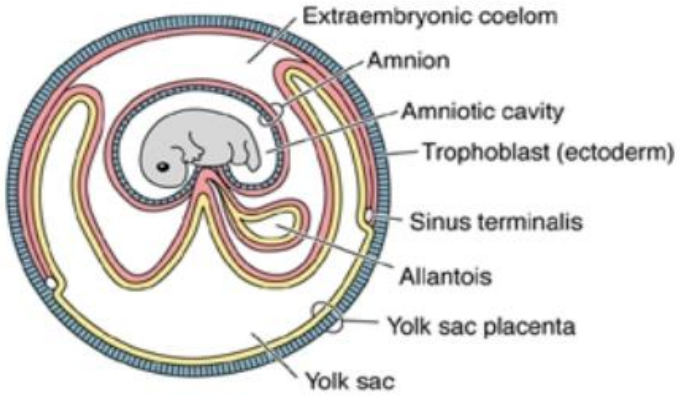
Thylacoleo carnifex

Owen 1871 *Phil Trans Roy Soc London* (161):213-66.

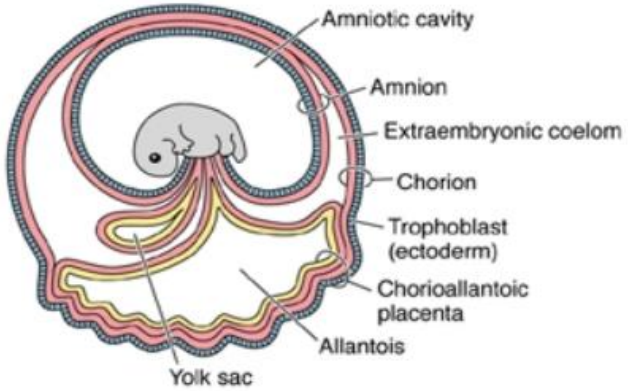
Placentas

- Marsupial placenta is adapted to reduce the gestation period
- Offspring therefore emerge less capable and less developed.
- Was once thought that the lack of marsupial diversity was due to this.
- However, peramelians have a convergent chorioallantoic placenta and birth larger young but are not diverse.
- It's more likely that marsupials can jettison their offspring in times of scarcity or environmental struggle, reducing the cost to the mother
- Australia, and south America for MOST of the Cenozoic were extremely variable and inhospitable environments to survive in.
- The cost of this strategy limits their developmental plasticity and thus their ability to evolve into new morphologies and new forms.

Marsupial mammals *do* have a placenta



Choriovitelline ("yolk sac") placenta of most marsupials (didelphids, paucituberculates, *Dromiciops*, notoryctids, diprotodonts, dasyuromorphs)



Chorioallantoic placenta of a placental mammal

Hyper-carnivory

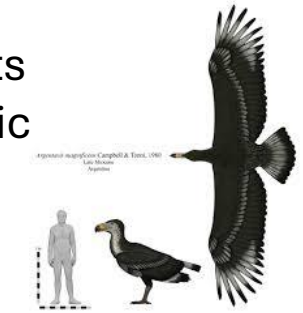
This is an interesting topic:

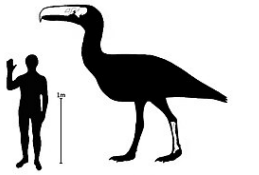
- Marsupial carnivores never get that big
- Also have weird adaptations not seen in placental carnivores
- Based on the literature I've read I would hypothesize that this is because Marsupial carnivores had to compete with larger reptilian carnivores, while placentals didn't.
- For example, Thylacosmilus' jaw and skull anatomy suggest it was better adapted to EATING quick rather than killing things.
- Similarly, Thylacoleo is mostly small with only 1 genus (carnifex) that got large.

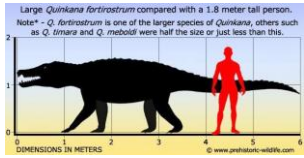
Terrestrial Reptilian competitors in Northern continents during late Cenozoic

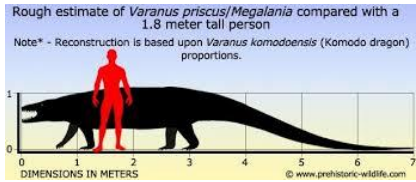
LITERALLY NOTHING

Terrestrial Reptilian competitors in southern continents during late Cenozoic









Placental hypercarnivory

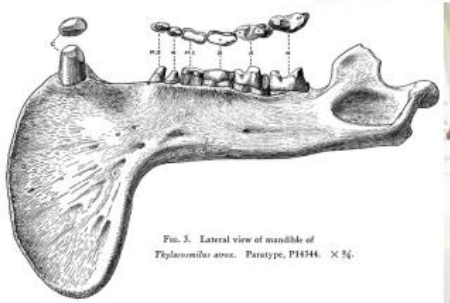
Smilodon fatalis

carnassials:
Smilodon: P4/m1
Thylacoleo: P3/p3

Thylacosmilus lacks true carnassials (see Janis et al. 2020 *PeerJ* https://doi.org/10.7717/peerj.9346)

Marsupial hypercarnivory

Thylacosmilus atrox



Thylacoleo carnifex

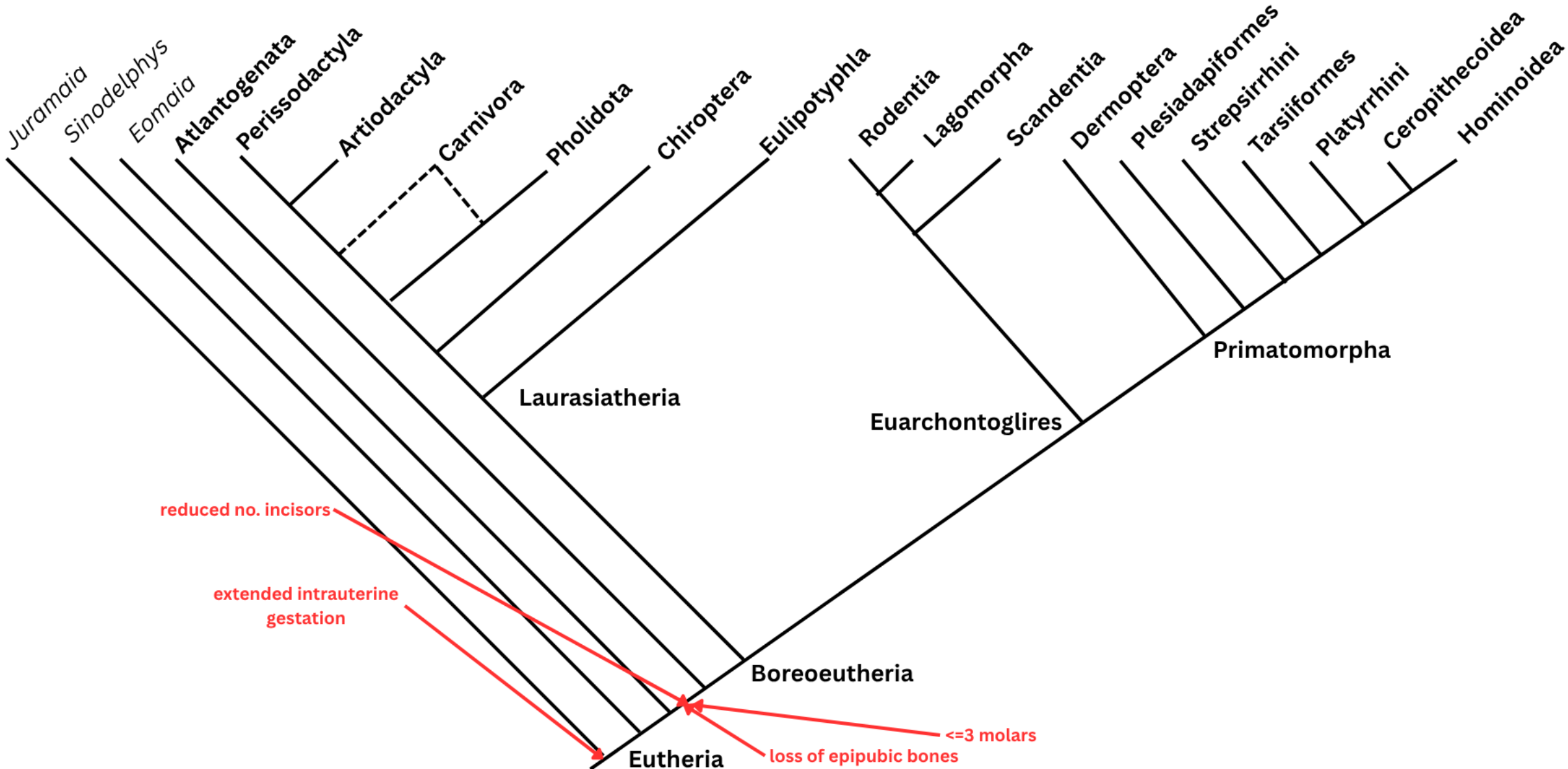
Notes

- Premolars and Molars are really hard to differentiate, it's virtually impossible to know what is M1, P4, P3, P2, and P1 in groups that have lost one or more of these teeth.
- Important thing is that they FUNCTION like those teeth.

Papers:

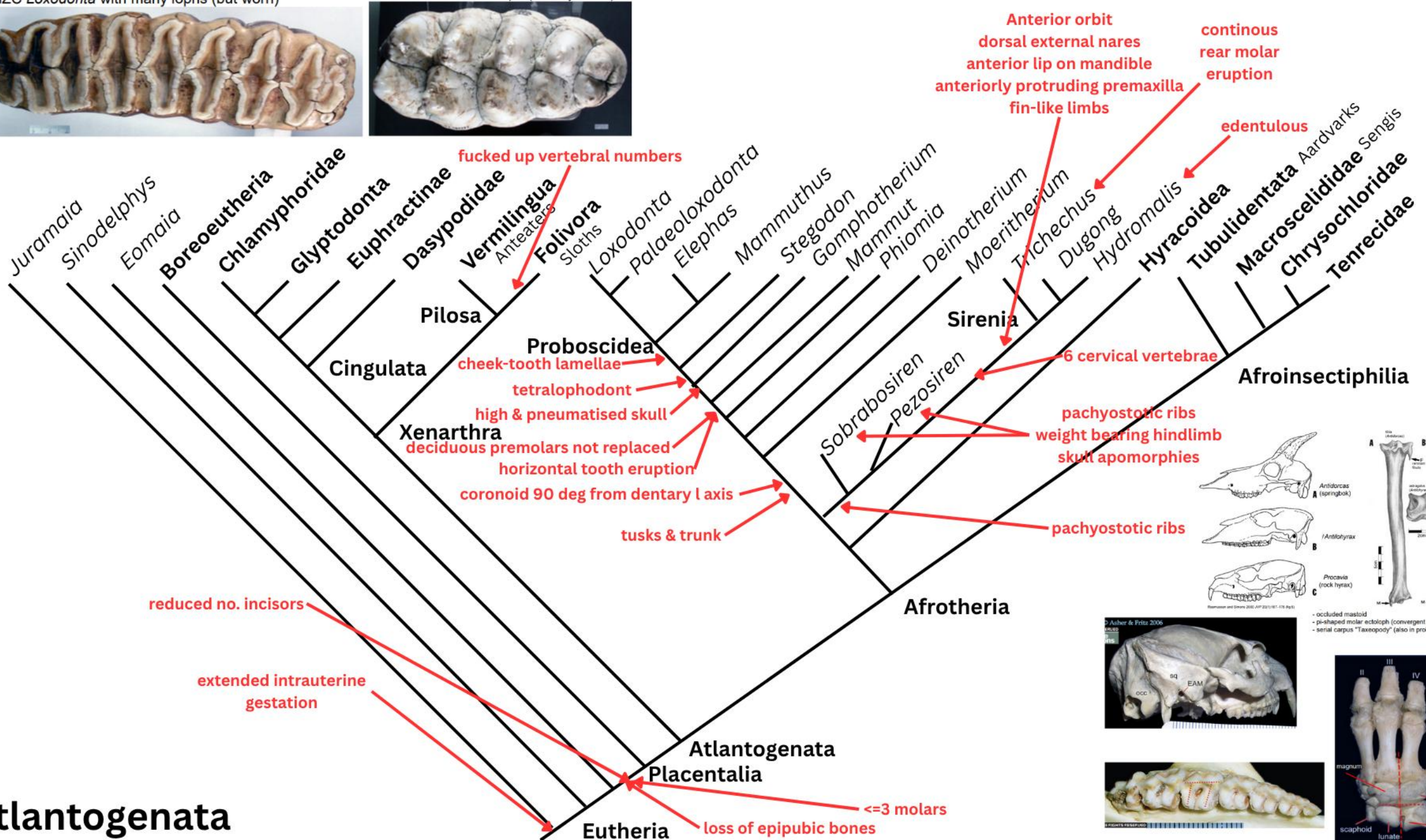
- Thylacosmilus, Thylacoleo, Smilodon tooth stress/ecology: C.M. Janis 2024, doi: 10.1002/ar.25444, Pollock et al. 2025, https://doi.org/10.1016/j.cub.2024.11.059

Boreoeutheria



UMZC *Loxodonta* with many lophs (but worn)

UMZC *Mammut* with fewer lophs (relatively unworn)



Atlantogenata

Shared derived traits

- Placentals have a few shared features, but because they're really diverse, they also have a MASSIVE number of disparate features.
- Extended intrauterine gestation
- Up to 3 molars
- Reduced no. incisors
- Lack epipubic bones

Evolution/Phylogeny

- Really messy phylogenetic tree before genetic methods
- Many groups have convergently evolved the same traits
- Many closely related groups have diverged really rapidly
- Makes it really hard to understand how groups are related to each other based on morphology alone.
- Even molecular evidence leaves some questions, for example there is a soft polytomy at the base of Placentalia with the relationship between Afrotheria, Xenartha, and Boreoeutheria being murky.
- Polytomies happen when groups diversify really quickly so it becomes hard to stitch back together the original relationship due to lack of fossil material and genetic support.
- Modern genetic evidence puts Xenartha and Afrotheria together with Atlantogenata as sister to Boreoeutheria but it doesn't have 100% support still.
- Other possibilities are Epitheria (Afrotheria + Boreoeutheria) and Exafroplacentalia (Xenartha + Boreoeutheria)

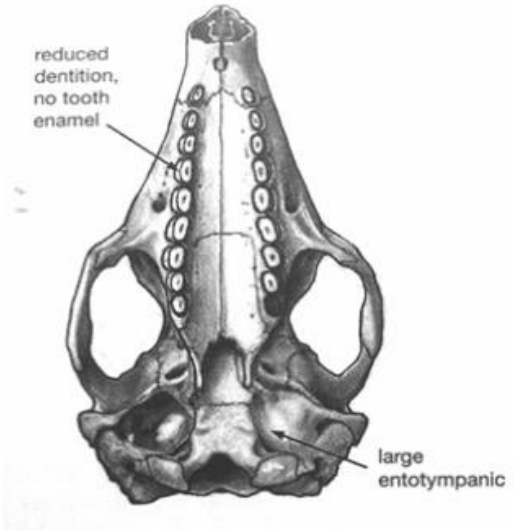
Xenarthra – **FUSED (SPINE) BONES FOR DIGGING**

- Deviations from typical vertebral numbers (cervical, thoracic, lumbar)
- This is conserved because of the diaphragm in parasagittal locomotion and the developmental constraints that follow.
- The developmental constraints are essentially that the diaphragm is innervated by tissues formed in the cervical region
- If this system is deviated from then it could mess up the formation and innervation and render the diaphragm non-functional

Xenarthra

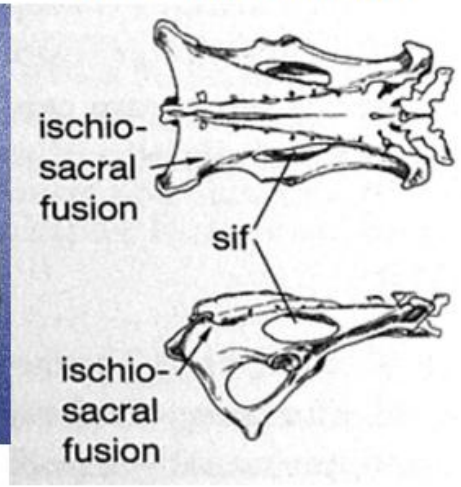
armadillos (29 spp.)
sloths (5 spp.)
anteaters (3 spp.)

Skull & teeth:
- no enamel
- entotympanic auditory bulla



Scapula:
- elongate acromion
- infraspinatus spine

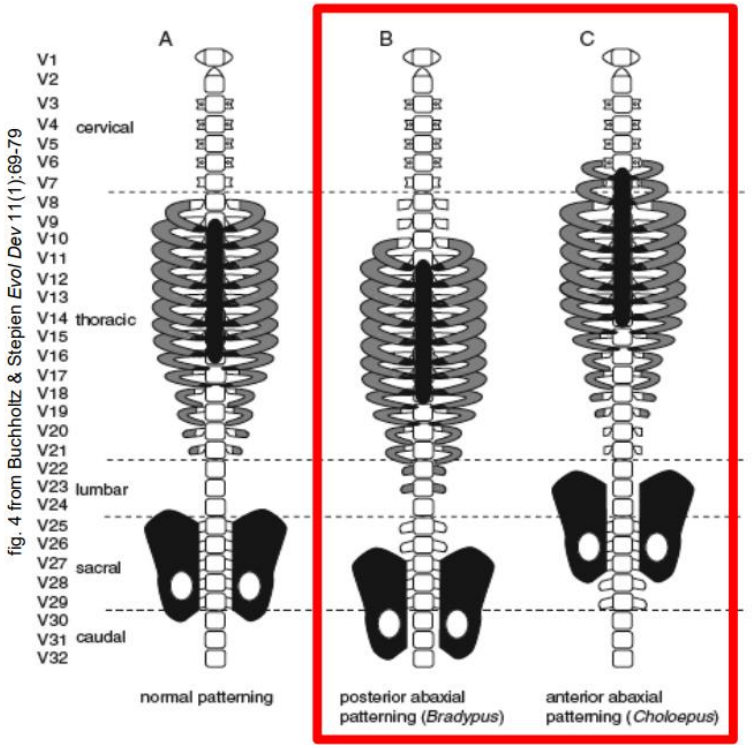
Pelvis:
- ischio-sacral fusion



Vertebrae:
- xenarthry

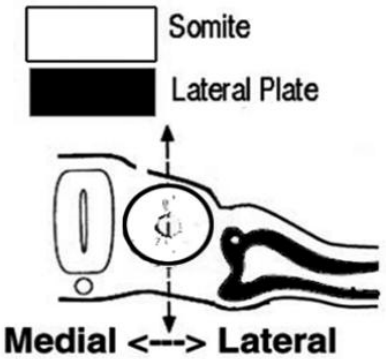
Deviation from vertebral column

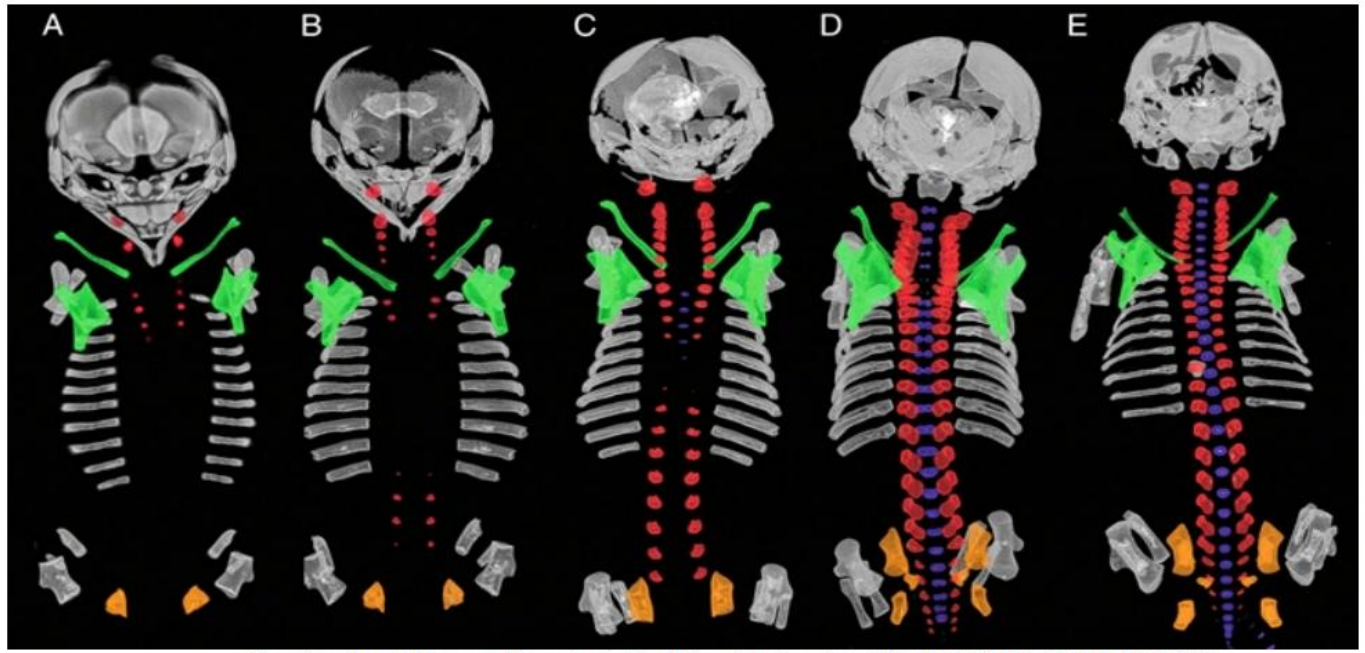
- No. cervicals & sacral position correlated
- Cervicals always ossify first
- This is still true in sloths, showing extra cervicals are ribless thoracics and vice versa
- One hypothesis is that the highly pleiotropic genes responsible for patterning cause risks via affecting other structures and increasing cancer risk.
- As manatees & sloths (in particular) have lower metabolism they can counteract this.
- However, there isn't sufficient evidence for this as most mammalian embryos don't show deviations from the pattern, xenarthrans do but only after the foetal stage – and non-endothermic synapsids still exhibit the constrained pattern.



Buchholtz & Stepien 2009
Evolution & Development 11(1):69–79
noted correlation between sacral position and number of neck vertebrae, and proposed Primaxial-Abaxial Shift hypothesis (PAS)

White: Primaxial (somitic) mesoderm
Black: Abaxial (lateral plate) mesoderm
Gray: uncertain in sloths





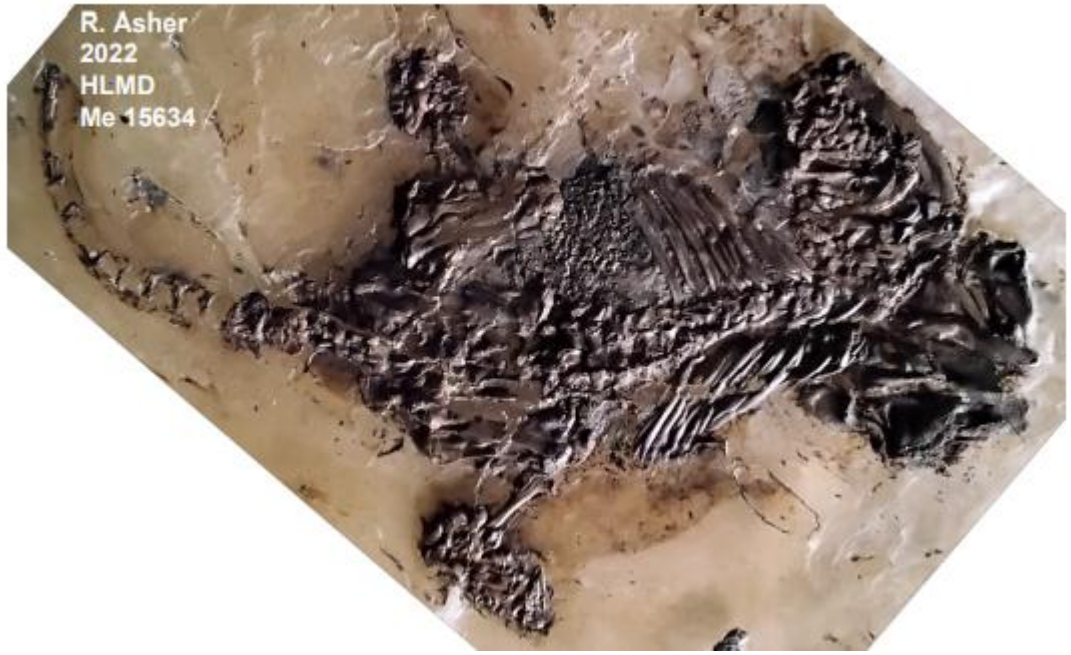
Growth series of *Dasypus* (Xenarthra). Fig. 1 from Hautier et al. 2010 *PNAS USA* 107(44):18903-08

Pangolins - Pholidota

- Exhibit extremely similar convergence with Cingulata.
- Lack xenarthry
- Scapula spine short + no infraspinatus spine
- Split terminal phalanges
- Elongate dorsally concave palate with separate pterygoid
- Loosely attached C-shaped premaxilla
- Glenoid “pedestal” for jaw articulation
- Anterior prong on anterior mandible
- Dermal scales

Eomanis waldi - eocene

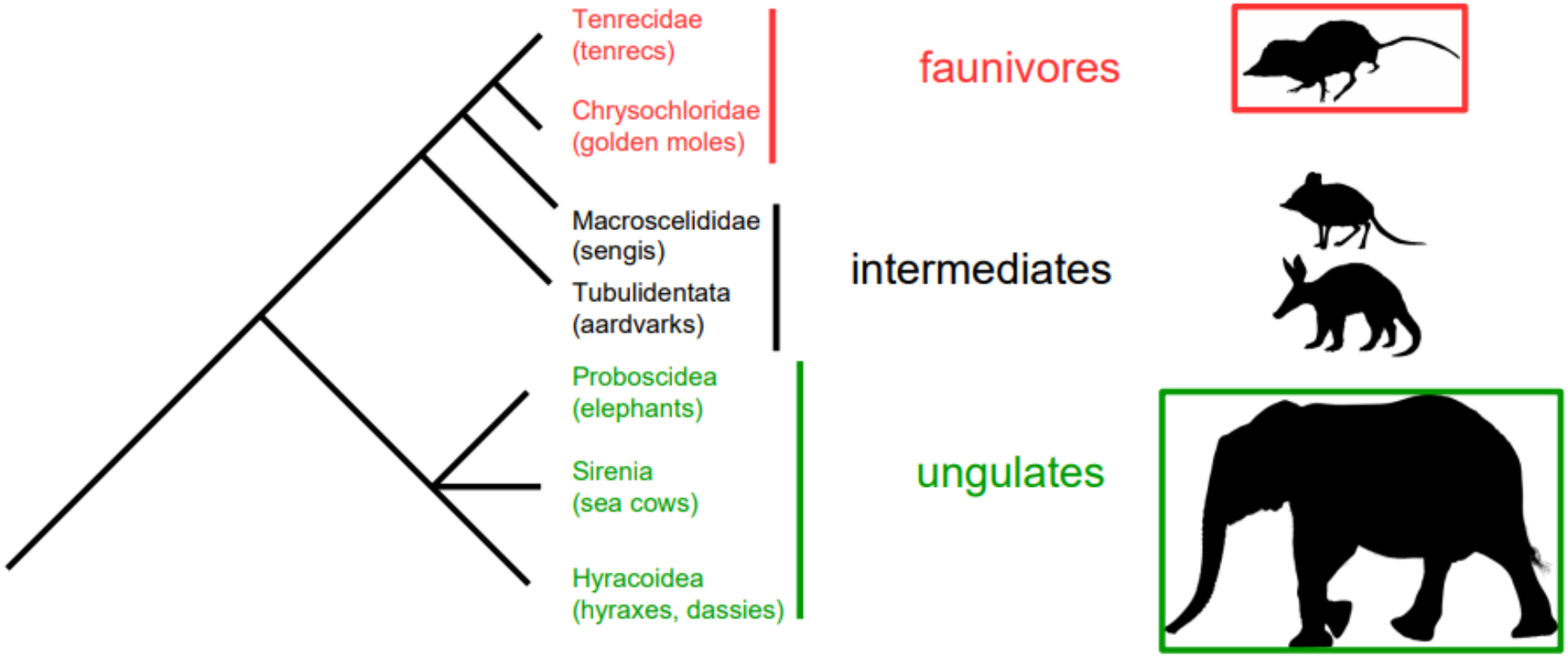
- Dermal scales
- Anterior prong
- No scapular coracoid process
- Edentulous
- No split phalanges
- Elongate scapular spine
- Scaphoid-lunate not fused



Eurotamandua joresi - eocene

- Lacks xenarthy
- Lacks elongate scapular spine and infraspinatus spine
- Ischial-sacrum articulation not fusion
- Probably a Pholidotan not a Cingulatan

Afrotheria



Ungulates:

- large body size
- herbivorous
- 4-cusped upper cheek teeth
- walk on tips of digits
- flattened terminal digits

Faunivores:

- small body size
- faunivorous
- 3-cusped upper cheek teeth
- walk on ankles/wrists
- narrow terminal digits

Sirenians

- Anteriorly situated orbit
- Dorsally situated external nares
- Anterior lip on mandible
- Anteriorly extensive premaxilla
- Pachyostotic ribs
- Molars erupt through life, horizontally (unless edentulous)

Species:

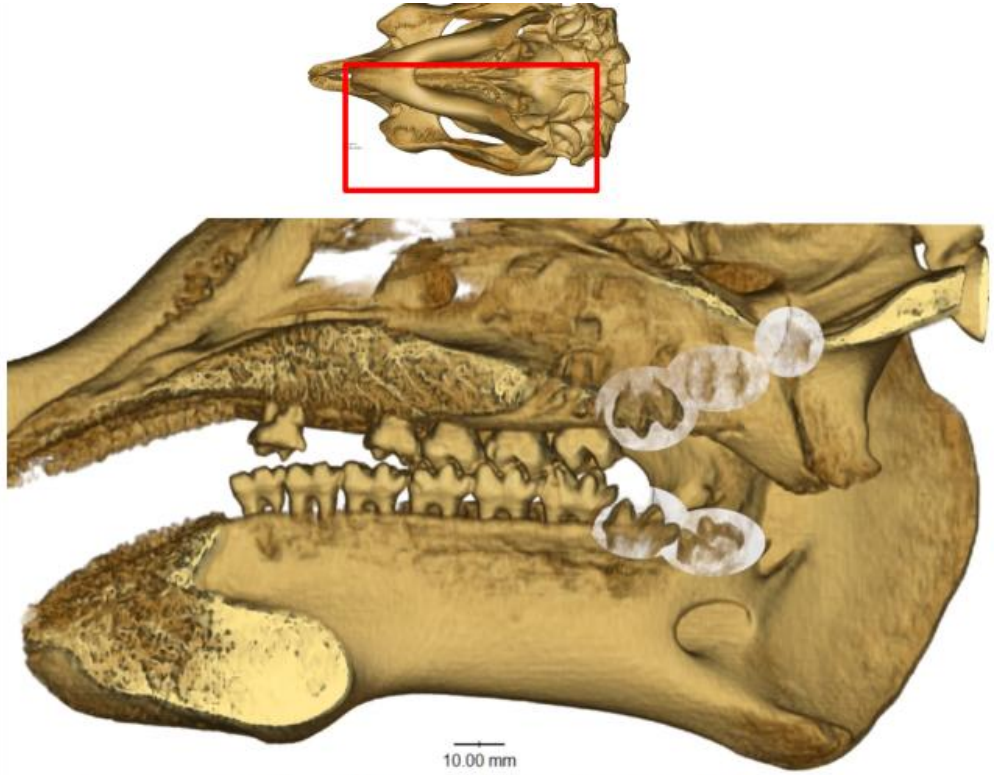
- Hydrodamalis gigas (Steller's sea cow – extinct)
- Dugong dugon (Dugong)
- Trichechus manatus (Manatee)

Evolutionary reasons for adaptations?

Pachyostosis?

Dorsal external nares?

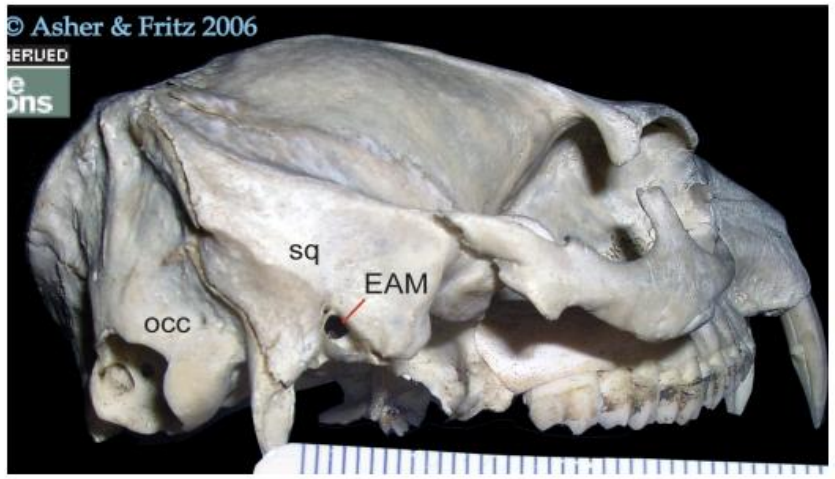
Continually erupting molars?



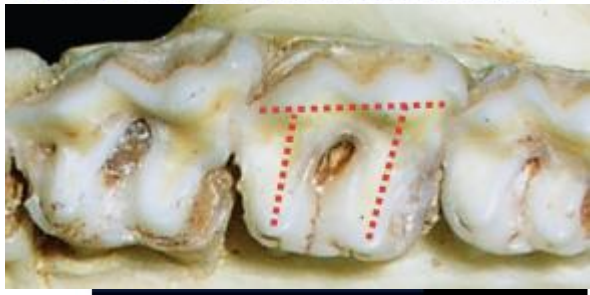
Trichechus senegalensis AMNH 53939 (http://digimorph.org/specimens/Trichechus_senegalensis/)
3D render by Robert J. Asher using Drishti 2.6.4 Limaye 2012, (https://doi.org/10.1117/12.935640)
and https://github.com/nci/drishti/

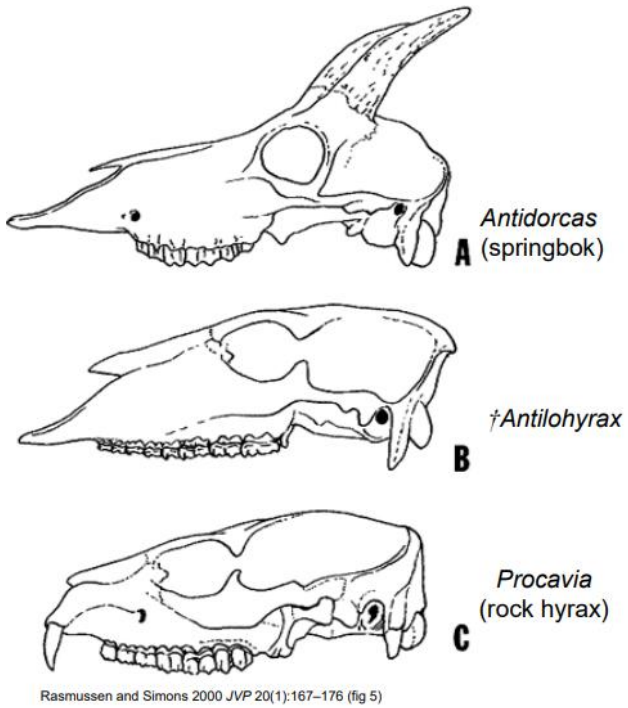
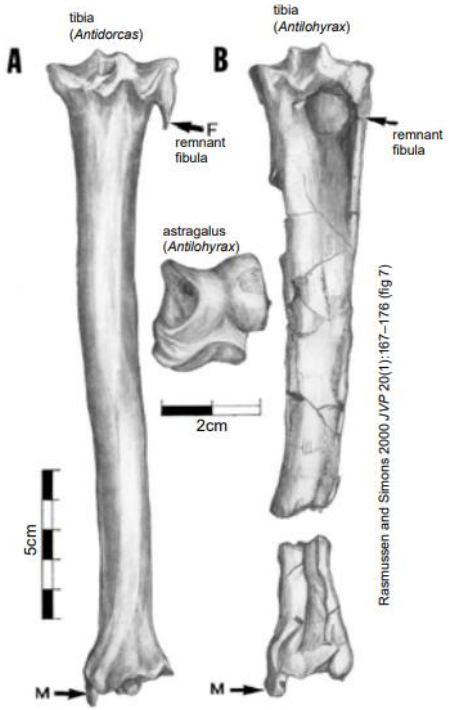
Hyracoidea

- Occluded mastoid
- Pi-shaped molar ectoloph (convergent with perissodactyls)
- Serial carpus (taxeopody – also in proboscideans)



Currently they exist as small generalist herbivores but during the Cenozoic they held niches similar to modern ungulates and often had convergent traits and anatomical features with them.



Extant Taxa:

- Dendrohyrax (3 spp.)
  - arboreal
- Heterohyrax (3 spp.) – yellow-spotted hyrax
- Procavia capensis – rock hyrax